StepFun Part I: Step Law, experiment by experiment

What was tuned, how optima were fitted, which quantities transfer, and how to reproduce the study

Question and position in the workflow

Li et al., Step Law v3 asks how **optimal learning rate and batch size** depend on model size and token horizon. It is a hyperparameter study, not a loss-allocation law. Part II, Farseer uses standardized HP choices when fitting a loss surface. Marin's tuning code offers an independent example of tuning a recipe before forecasting with it.

Local primary files: PDF, metadata/LaTeX availability, and released-table audit. Official fitting artifacts come from step-law/steplaw, pinned in the local archive. Distinguish released tabular observations from all experiments counted in the paper.

Fix the scientific object before reading the coefficients

The study's conditional objective is

$$(\eta_\star, B_\star) = \arg\min_{\eta,B} L(\mathcal{A}, \mathcal{D}, N, D, \eta, B),$$

where $\mathcal{A}$ is the architecture family, $\mathcal{D}$ the data recipe, N its declared parameter count, D training tokens, η peak LR and B token batch. The paper's §1.1 distinguishes parameter conventions and Table 5 reports total and active MoE counts. Our stored/active/executed counters must not be substituted interchangeably. In the released CSVs, `N` and `Na` are separate columns; inspect the counting code and exact family before evaluating the predictor.

Reported setup (§3.2): BPE vocabulary 65,536; context 2,048; RMSNorm/SwiGLU; principally ALiBi; AdamW $\beta_1 = 0.9$, $\beta_2 = 0.95$, $\epsilon = 10^{-8}$, decay 0.1 and clip 1; 2,000 warmup updates; cosine decay to an **absolute** final LR 10^{-5}. This final-LR convention is part of the experiment: fixing a peak/final ratio instead can shift the optimum.

1. Search the LR/batch surface

The LR grid uses powers of two with half-exponent spacing from $2^{-10.5}$ through 2^{-7}. The batch grid spans 32,768–4,194,304 **tokens**, doubling between values. Figure 1 describes a 120-run LR/batch comparison for a 1B/100B-token model; other figures examine different sizes/horizons.

The practical observation is a bowl/plateau in the tested two-dimensional HP landscape. This is empirical local structure, not a theorem that every optimizer/model has a globally convex loss surface. A grid point can be unstable or boundary-limited. Fit only supported local minima; retain the raw contour and failed runs.

Figure 3 compares smoothed training loss with held-out loss in a controlled setting. Their optimal settings coincide in the shown comparison. This supports smoothed loss as a cheap proxy **there**, not comparing training losses from different data mixtures. Our mixture experiments need one fixed held-out target distribution.

2. Check schedule dependence

The paper compares an absolute final LR against a final LR tied to peak. When peak changes, a ratio-based endpoint changes too; the search is no longer varying only the early/mid-training rate. A reproduction must preserve both warmup and annealing semantics. Compare final annealed losses, not a snapshot at different schedule phases.

This connects to Marin's constant-versus-decaying LR follow-up, where the useful multiplier bracket changes. A transfer law is conditional on its schedule and optimizer groups.

3. Fit a law in size and horizon

The reported fitted form (Eq.1/Table 2) is

$$\eta_\star = 1.79 N^{-0.713} D^{0.307}, \qquad B_\star = 0.58 D^{0.571}.$$

Here the published numerical coefficients expect raw parameter/token units and return token batches. They are not valid after changing N from parameters to billions without converting the intercept.

The fitting uses log-linear regressions,

$$\log \eta_\star = \log c + \alpha \log N + \beta \log D, \qquad \log B_\star = \log d + \gamma \log D,$$

with 1,000 bootstrap fits. The experiment description spans seven model scales and multiple token horizons and reserves larger-size/horizon settings for checks. This variation is what distinguishes an HP surface from a fixed-TPP ray.

Under the published equation, doubling N at fixed D multiplies peak LR by about 0.61. Increasing D fourfold while co-selecting batch multiplies LR by about 1.53 and batch by about 2.21. These are mathematical consequences of this fit, not new measured results. A positive joint-optimum horizon exponent does not imply that LR rises with horizon when batch and schedule are held fixed under another recipe.

4. Test architecture and data transfer

Table 4 includes width 960/depth7, width 1280/depth10 and width 2048/depth16. Thus the paper itself does not require power-of-two widths or fixed depth. Table 5 covers sparse models with differing top-k/active counts; Table 6 varies code/math proportions and includes English/Chinese data.

The reported transfer results justify testing whether a law survives moderate changes, not declaring every new mixture requires a complete search. They also do not certify a recurrent MoE at trillions of tokens. Our target's attention, tokenizer, optimizer, router, horizon and expert geometry differ.

What is actually in the public fitting archive?

At the inspected official-code revision, we counted **1,911 dense rows**, **708 MoE rows** and **1,000 bootstrap coefficient rows**. The paper's headline describes roughly 3,700 trained models; those are different scopes. The release is not evidence that we possess every original training log/checkpoint.

`dense_lr_bs_loss.csv` stores shape, LR, `bs`, updates, smooth loss, N, D and TPP. `moe_lr_bs_loss.csv` also stores sequence length, routing/expert details, active count, compute and runtime fields. **CSV `bs` is sequences**; multiply by context length (2,048 for the studied release) before comparing it with Eq.1's token batch. A descriptive (N, D) grouping

can still mix shapes/recipes; use full configuration keys when fitting a causal comparison.

`code/fit_tool.py` provides `pred-opt-lr-bs N D seq_len`. It is useful for independent numerical checks and suggested search centers. Do not promote its output to a trainer config without confirming count conventions, schedule and group-wise LR definitions.

A minimal replication protocol for our team

1. Freeze one standard architecture/data/tokenizer/optimizer/schedule and an independent fixed validation set.
2. Recreate two or three released LR/batch cells with their actual settings, including absolute final LR and sequence/token batch conversion. Check contour shape and metric agreement.
3. Fit local optima with interior minima, distinguish observed from interpolated losses and retain censored/unstable cells.
4. Expand to independently varying widths and horizons; fit the log-linear law and compare a simpler batch-only-in-D law against a size-dependent alternative.
5. Evaluate unseen size and longer-horizon cells with local brackets. For our heavy-overtraining goal, include 200+ **stored** TPP checks and report the corresponding active TPP.
6. Run a small cyber-mixture transfer check before refitting everything. A new optimizer, router or loop architecture requires separate qualification.

The heavy tuning plan sets the concrete grid/BO stages and explains how to fund them. No published coefficient is treated as an empirically selected setting for our hero run.